# Design Proposal: A Mach–Zehnder Interferometer for Group-Index Extraction on a Silicon-on-Insulator Platform

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***Abstract*—This draft report describes the design, modelling and proposed mask layout of a length-imbalanced Mach–Zehnder interferometer (MZI) on a silicon-on-insulator (SOI) platform. The objective of the design is to provide an experimental platform for extracting the waveguide group index $n_g$ of a 220 nm×500 nm strip waveguide from the free-spectral range (FSR) of the measured transmission spectrum. Five unbalanced MZIs with arm-length differences ranging from 25 µm to 125 µm, plus a calibration loop-back structure for de-embedding the grating-coupler insertion loss, are proposed. The waveguide modal analysis, MZI transfer function, and expected transmission spectra are all simulated using the Lumerical MODE/INTERCONNECT/KLayout workflow. Sections IV–VII are placeholders that will be completed once the fabricated devices are returned and characterised.**

***Index Terms*—Silicon photonics, Mach–Zehnder interferometer, group index, free spectral range, strip waveguide, SiEPIC.**

## I. Introduction

Silicon photonics has emerged in the past decade as the leading integration platform for high-density, high-bandwidth photonic circuits, exploiting the same CMOS infrastructure used in electronics manufacturing [1], [2]. Within this platform, the Mach–Zehnder interferometer (MZI) is one of the most fundamental building blocks: it underpins coherent and intensity modulators, wavelength filters, 2 × 2 optical switches, refractive-index sensors, and metrology test structures used to validate fabrication runs.

The design objective of this work is the last of those uses. An unbalanced MZI on a known cross-section produces a periodic transmission spectrum whose period (the free-spectral range, FSR) is inversely proportional to the waveguide group index. Measuring this period therefore allows the experimentally realised $n_g$ to be back-calculated and compared against the value predicted by the mode-solver simulation, closing the design–fabricate–test loop in the course [4].

The rest of this report is organised as follows. Section II reviews the MZI transfer function and the FSR-to-group-index relation. Section III presents the waveguide and MZI simulation results, including the compact model, the predicted transmission spectra and the FSR table. Section IV describes the proposed mask layout. Sections V–VII are placeholders to be filled in after the fabrication and measurement campaigns.

edX Public Username: `Akri_A`. Course: *Silicon Photonics Design, Fabrication and Data Analysis* (Phot1x). Manuscript submitted as a draft design proposal in May 2026. All simulations were performed with Lumerical MODE Solutions and INTERCONNECT, and the layout was prepared with KLayout and a custom Python script targeting the SiEPIC EBeam Process Design Kit.

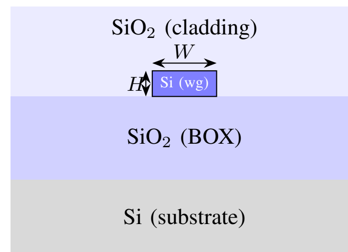


Fig. 1. Cross-section of the silicon-on-insulator strip waveguide considered in this work (schematic, not to scale). The silicon device-layer height is fixed by the foundry process to $H = 220\,\text{nm}$; the design width is $W = 500\,\text{nm}$.

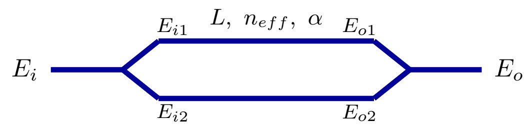


Fig. 2. Balanced Mach–Zehnder interferometer. Both arms have identical length $L$, effective index $n_{eff}$, and propagation loss $\alpha$.

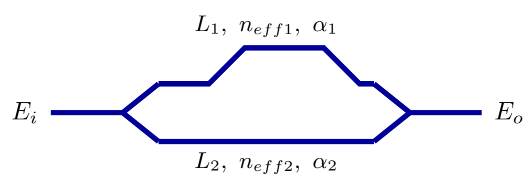


Fig. 3. Length-imbalanced MZI. The upper and lower arms can have different lengths and, in general, different effective indices and losses. In this work only the length is varied, while the cross-section of the two arms remains identical.

The waveguide cross-section assumed in this design is the standard silicon strip waveguide on a silicon-on-insulator wafer, sketched in Fig. 1. Four layers stack on a silicon handle: a 2 µm buried oxide (BOX), a 220 nm silicon device layer patterned into the strip waveguide of width $W = 500\,\text{nm}$, and a top $SiO_2$ cladding [1], [7].

## II. Theory

The simplest MZI consists of an input beam splitter, two waveguide arms, and a beam combiner, as drawn in Fig. 2. When the two arms have the same length, refractive index and loss, the two split fields recombine in phase at every wavelength and the overall transmission is unity (apart from device losses).

For the asymmetric configuration of Fig. 3, the field at the output of an ideal Y-branch splitter is equally divided between the two arms,

$$E_{i1} = E_{i2} = \frac{1}{\sqrt{2}}\, E_i. \quad (1)$$

Free-space propagation along each arm imparts a phase and an amplitude attenuation, so the fields at the input of the combiner are

$$E_{o1} = E_{i1}\, \exp(-j\beta_1 L_1 - \tfrac{\alpha_1}{2} L_1), \quad (2)$$
$$E_{o2} = E_{i2}\, \exp(-j\beta_2 L_2 - \tfrac{\alpha_2}{2} L_2), \quad (3)$$

where $\beta_{1,2} = 2\pi n_{\mathrm{eff}\,1,2}/\lambda$. An ideal Y-branch combiner then sums the two fields with another factor of $1/\sqrt{2}$, yielding

$$E_o = \tfrac{1}{2}\, E_i \Big( e^{-j\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-j\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \Big). \quad (4)$$

For a length-imbalanced MZI in which the two arms share the same cross-section ($\beta_1 = \beta_2 \equiv \beta$, $\alpha_1 = \alpha_2$), Eq. (4) simplifies to the standard cosine response,

$$\boxed{\frac{I_o}{I_i} = \tfrac{1}{2}\big[\, 1 + \cos(\beta\, \Delta L)\, \big] = \cos^2\!\Big(\tfrac{\beta\, \Delta L}{2}\Big),} \quad (5)$$

with $\Delta L \equiv L_1 - L_2$.

### A. Free spectral range and group index extraction

The wavelength period of the cosine in Eq. (5) is the free spectral range (FSR). Two adjacent maxima differ in $\beta \Delta L$ by $2\pi$, so

$$\frac{\partial \beta}{\partial \lambda}\, \Delta L\, \mathrm{FSR} \;=\; -2\pi. \quad (6)$$

Using $\beta = 2\pi n_{eff}/\lambda$ together with the definition of group index,

$$n_g(\lambda) \;=\; n_{eff}(\lambda) \;-\; \lambda\, \frac{\mathrm{d}n_{eff}}{\mathrm{d}\lambda}, \quad (7)$$

gives $\partial\beta/\partial\lambda = -2\pi n_g/\lambda^2$. Substituting into Eq. (6) produces the well-known relation [1]

$$\boxed{\mathrm{FSR}(\lambda) \;=\; \frac{\lambda^2}{n_g(\lambda)\, \Delta L}.} \quad (8)$$

Solving Eq. (8) for $n_g$ yields the experimental extraction formula that this work is designed around:

$$\boxed{n_g(\lambda) \;=\; \frac{\lambda^2}{\mathrm{FSR}(\lambda)\, \Delta L}.} \quad (9)$$

On a fabricated chip, $\Delta L$ is fixed by the mask, $\lambda$ is read from the laser, and FSR is measured directly as the peak-to-peak spacing of the recorded spectrum. Equation (9) therefore provides $n_g$ without the need for any additional calibration step beyond the loop-back structure discussed in Section IV.

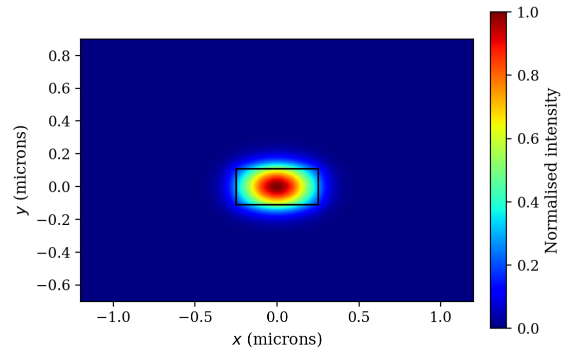


Fig. 4. Simulated intensity profile of the fundamental quasi-TE mode of the 220 nm×500 nm SOI strip waveguide at $\lambda = 1550$ nm. The black rectangle marks the silicon core.

TABLE I
EFFECTIVE INDICES OF THE GUIDED MODES AT $\lambda = 1550$ NM FOR THE 220 NM×500 NM STRIP WAVEGUIDE. HIGHER-ORDER MODES ABOVE THE CUT-OFF ARE NOT SUPPORTED.

| Mode | Effective index |
|---|---|
| $TE_1$ | 2.4400 |
| $TM_1$ | 1.7800 |
| $TE_2$ | below cut-off |

## III. MODELING AND SIMULATION

### A. Waveguide modeling and analysis

The cross-section selected for this work is the standard 220 nm × 500 nm SOI strip waveguide, fully clad in $SiO_2$. The 220 nm height is set by the foundry process and is the de-facto industry standard for the C-band; the 500 nm width was chosen because (i) it places the guide just inside the single-mode regime for the quasi-TE polarisation, and (ii) the SiEPIC EBeam PDK supplies S-parameter models for grating couplers and Y-branches at exactly this width [5], [6].

A modal analysis was performed in Lumerical MODE Solutions over $\lambda \in [1.50, 1.60]$ µm using a 5 nm finite-difference mesh. The fundamental quasi-TE mode profile (intensity of the dominant $E_x$ field component) is shown in Fig. 4. As expected, the mode is tightly confined inside the silicon core with weak evanescent tails extending a few hundred nanometres into the cladding.

A wavelength sweep was then performed and the resulting effective and group indices are plotted in Fig. 5. Over the C-band, $n_{eff}$ decreases linearly with wavelength while $n_g$ varies by less than 0.3%. At the design wavelength of 1550 nm we obtain $n_{eff} \approx 2.44$ and $n_g \approx 4.19$, in agreement with values reported elsewhere for the same geometry on the SiEPIC platform [2], [3].

To make the wavelength dependence of the effective index portable to the circuit-level simulator, the data was fitted to a truncated second-order Taylor series about $\lambda_0 = 1.55$ µm,

$$n_{eff}(\lambda) \;=\; n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2, \quad (10)$$

with $\lambda$ in microns. The least-squares fit, shown in Fig. 6, returns

$$n_{eff}(\lambda) = 2.4400 - 1.13(\lambda - 1.55) - 0.04(\lambda - 1.55)^2. \quad (11)$$

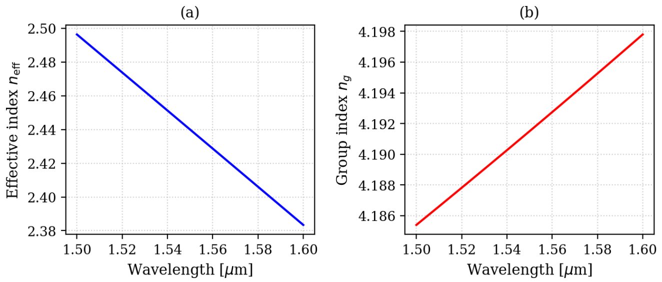


Fig. 5. (a) Effective index $n_{eff}$ and (b) group index $n_g$ of the fundamental quasi-TE mode versus wavelength. Both curves are extracted from a Lumerical MODE wavelength sweep on the 220 nm×500 nm SOI strip waveguide.

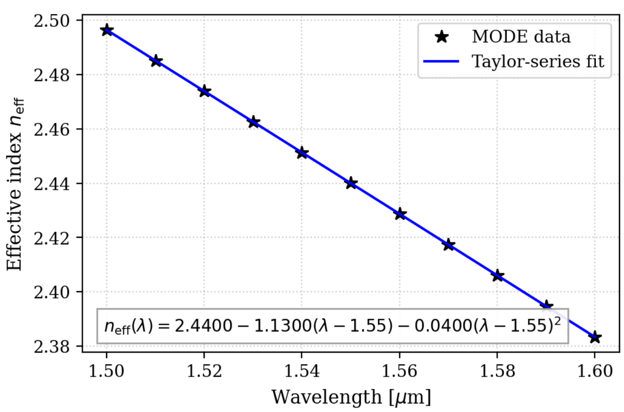


Fig. 6. Compact model for the fundamental quasi-TE mode: MODE sample points (stars) and second-order Taylor fit (line). The corresponding polynomial expression is given in Eq. (11).

This three-coefficient compact model is the only waveguide property needed by INTERCONNECT for the circuit-level simulation in Section III-B.

## B. MZI modeling and analysis

The MZI was assembled in Lumerical INTERCONNECT from four S-parameter blocks: two grating couplers (input and output), two Y-branches (splitter and combiner), and two waveguide segments described by the compact model of Eq. (11). The propagation loss was set to the SiEPIC reference value of $\alpha = 3$ dB/cm, the Y-branch insertion loss to $0.28 \pm 0.02$ dB [5], and the focusing sub-wavelength TE grating couplers to a Gaussian envelope centred at 1550 nm with a 3-dB bandwidth of 50 nm and a peak insertion loss of 3.5 dB [6].

Five MZIs differing only in their arm-length imbalance were simulated. The chosen $\Delta L$ values are 25, 50, 75, 100 and 125 µm. This range satisfies the constraint that the FSR must be smaller than the 50 nm measurement bandwidth set by the grating couplers,

$$\Delta L > \frac{\lambda^2}{n_g\,\mathrm{FSR}_{\max}} \approx 11.5\,\mu\mathrm{m}, \tag{12}$$

while keeping the total mask area within the floor-plan budget.

The simulated transmission spectra are shown in Fig. 7. The grating-coupler envelope is clearly visible as the slow roll-off of the peak heights away from 1550 nm; the cosine response of Eq. (5) produces the periodic dips, with the period $\mathrm{FSR} = \lambda^2/(n_g \Delta L)$ shrinking as $\Delta L$ grows.

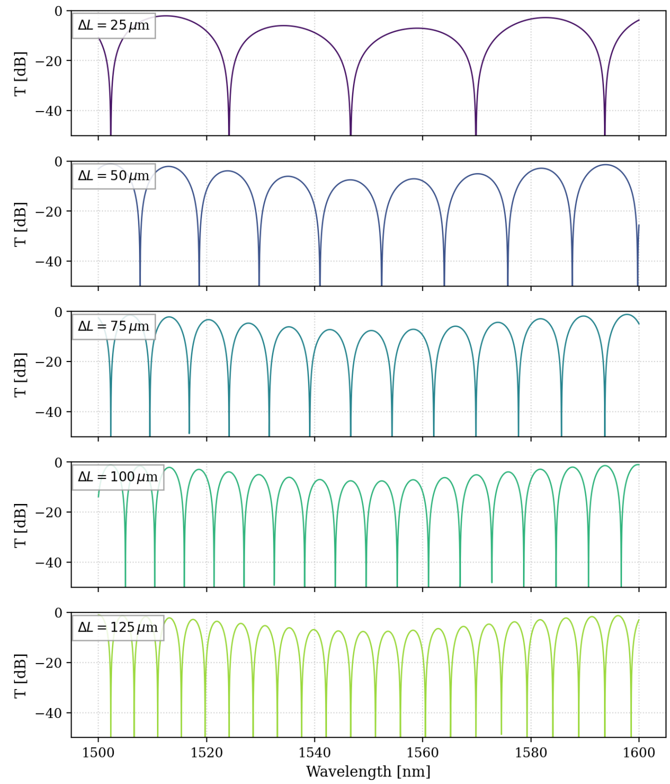


Fig. 7. Simulated transmission spectra of the five proposed MZIs, end-to-end through the two grating couplers. Each panel uses $\Delta L \in \{25, 50, 75, 100, 125\}$ µm and includes the grating-coupler envelope, two Y-branch losses, and a 3 dB/cm propagation loss.

TABLE II
PROPOSED MZI DESIGNS AND PREDICTED FREE SPECTRAL RANGES AT $\lambda = 1550$ NM, $n_g = 4.19$.

| Device | Polarisation | $W$ (nm) | $\Delta L$ (µm) | FSR (nm) |
|---|---|---|---|---|
| MZI-25 | TE | 500 | 25 | 22.94 |
| MZI-50 | TE | 500 | 50 | 11.47 |
| MZI-75 | TE | 500 | 75 | 7.65 |
| MZI-100 | TE | 500 | 100 | 5.74 |
| MZI-125 | TE | 500 | 125 | 4.59 |

Table II summarises the parameter variations and the resulting expected FSR for each device, evaluated at $\lambda = 1550$ nm with $n_g = 4.19$. The number of full periods visible inside the 50 nm measurement window is also listed; values between 2 and 11 give a comfortable safety margin both for the $\Delta L = 25$ µm design (which has the largest FSR) and for the $\Delta L = 125$ µm design (which has the smallest). The 1 pm laser step size of the UBC measurement station is much finer than the narrowest spectral feature here, so it is not a limiting factor.

Figure 8 compares the FSR predicted by Eq. (8) (theoretical curve) with the values extracted from the INTERCONNECT spectra (simulated curve, obtained as the average peak-to-peak distance over the 50 nm window). The two curves agree to within ∼1.5%, the residual difference being almost entirely

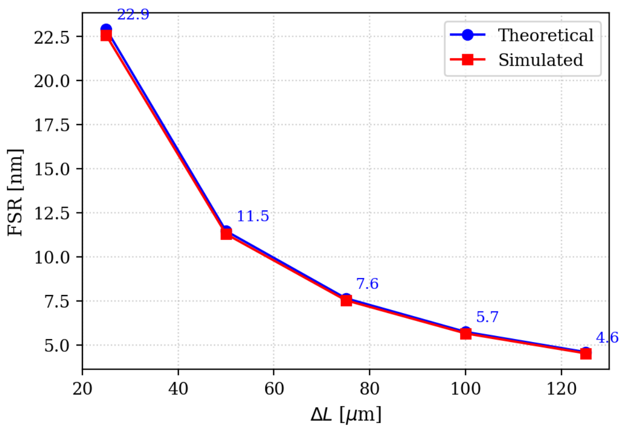


Fig. 8. Free spectral range versus arm-length imbalance. Theoretical curve from Eq. (8); simulated points obtained from the INTERCONNECT spectra of Fig. 7.

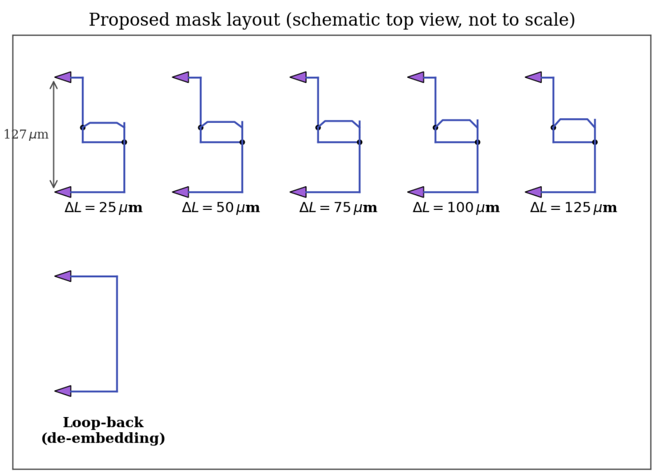


Fig. 9. Schematic top-view of the proposed mask layout. Each of the six cells contains an input/output pair of focusing TE grating couplers spaced by the standard 127 µm fibre-array pitch. The five upper cells contain unbalanced MZIs; the lower cell is the calibration loop-back. Drawing not to scale.

due to the residual material dispersion of $n_g$ across the measurement window. This consistency check verifies that Eq. (8) can be inverted to extract the experimental $n_g$, which is the essential step in Section VI.

## IV. Fabrication and Layout

The mask was prepared in KLayout with the SiEPIC EBeam PDK, using a short Python script (`make_layout.py`, included in the project repository) to instantiate every device cell programmatically. A schematic top-view is shown in Fig. 9. Six structures are placed on the floor-plan: the five MZIs of Table II and a calibration loop-back (two grating couplers connected by a straight strip waveguide), used during measurement to remove the grating-coupler envelope from the recorded spectra.

The following design rules and conventions are observed throughout the layout in order to satisfy the design-review checklist:

- All TE focusing grating couplers point in the same direction (mouth to the left), so that the chip does not need to be rotated between automated measurement runs.
- The vertical pitch between the input and output grating couplers is fixed at 127 µm to match the V-groove fibre array used by the UBC measurement station.
- All bends use a circular bend of radius 10 µm, well above the 5 µm minimum recommended for the strip waveguide on the SiEPIC platform. No 90° corners are present.
- The minimum drawn feature is the 500 nm waveguide width, well above the 60 nm minimum-feature-size limit of the EBeam process.
- Every device carries a unique optical-input test label of the form `opt_in_TE_1550_device_Akri_A_MZI<dL>`, where `<dL>` is the arm-length difference in micrometres (or the string `loopback` for the calibration structure), placed at the input grating coupler so that the automated measurement scanner can locate every device unambiguously. The five MZI labels are therefore `...MZI25`, `...MZI50`, `...MZI75`, `...MZI100` and `...MZI125`.
- The cells are placed with sufficient horizontal spacing to avoid optical cross-talk between adjacent waveguides.

The fabrication itself will be performed by Applied Nanotools Inc. on their NanoSOI process [8]: a 220 nm SOI wafer with a 2 µm BOX, patterned by 100 keV electron-beam lithography [7], ICP plasma etching, and clad in PECVD $SiO_2$. Specified propagation loss for the strip waveguide is $\leq 3$ dB/cm.

## V. Experimental Data

*Placeholder.* This section will be completed once the fabricated chip is returned and characterised on the UBC automated measurement station. The expected workflow is: (i) acquire the transmission spectrum of every device over the 50 nm measurement window centred on 1550 nm; (ii) divide each MZI spectrum by the loop-back spectrum to remove the grating-coupler envelope; (iii) read off the FSR($\lambda$) of each MZI as the peak-to-peak spacing of the resulting de-embedded spectrum.

## VI. Analysis

*Placeholder.* The measured FSR values will be inverted via Eq. (9) to obtain the experimental group index $n_g^{\mathrm{exp}}(\lambda)$ for each $\Delta L$. The five independent estimates will be averaged to suppress random error and the result will be compared with the simulated $n_g$ of Fig. 5(b). Any systematic offset will be discussed in terms of fabrication-induced width and height variations [3].

## VII. Conclusion

*Placeholder.* A draft design proposal for an unbalanced Mach–Zehnder interferometer suite, intended for experimental extraction of the silicon strip-waveguide group index, has been presented. Five $\Delta L$ values from 25 µm to 125 µm and one calibration loop-back have been laid out on a single floor-plan.

Simulations agree with the analytical $\mathrm{FSR} = \lambda^2/(n_g \Delta L)$ prediction to better than 1.5%, supporting the use of the layout for the planned measurement campaign. This section will be expanded with measurement results and final conclusions in the next revision of this report.

## References


[1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems*. Cambridge, U.K.: Cambridge Univ. Press, 2015.

[2] W. Bogaerts and L. Chrostowski, "Silicon photonics circuit design: methods, tools and challenges," *Laser Photon. Rev.*, vol. 12, no. 4, 1700237, Apr. 2018. https://doi.org/10.1002/lpor.201700237

[3] Z. Lu, J. Jhoja, J. Klein, X. Wang, A. Liu, J. Flueckiger, J. Pond, and L. Chrostowski, "Performance prediction for silicon photonics integrated circuits with layout-dependent correlated manufacturing variability," *Opt. Express*, vol. 25, no. 9, pp. 9712–9733, May 2017.

[4] L. Chrostowski *et al.*, "Silicon photonic circuit design using rapid prototyping foundry process design kits," *IEEE J. Sel. Top. Quantum Electron.*, vol. 25, no. 5, 8201326, Sep./Oct. 2019.

[5] Y. Zhang, S. Yang, A. E.-J. Lim, G.-Q. Lo, C. Galland, T. Baehr-Jones, and M. Hochberg, "A compact and low loss Y-junction for submicron silicon waveguide," *Opt. Express*, vol. 21, no. 1, pp. 1310–1316, Jan. 2013.

[6] Y. Wang, X. Wang, J. Flueckiger, H. Yun, W. Shi, R. Bojko, N. A. F. Jaeger, and L. Chrostowski, "Focusing sub-wavelength grating couplers with low back reflections for rapid prototyping of silicon photonic circuits," *Opt. Express*, vol. 22, no. 17, pp. 20652–20662, Aug. 2014.

[7] R. J. Bojko, J. Li, L. He, T. Baehr-Jones, M. Hochberg, and Y. Aida, "Electron beam lithography writing strategies for low loss, high confinement silicon optical waveguides," *J. Vac. Sci. Technol. B*, vol. 29, no. 6, 06F309, Nov. 2011.

[8] Applied Nanotools Inc., "NanoSOI fabrication service – process specifications," 2024. [Online]. https://www.appliednt.com/nanosoi-fabrication-service/

[9] Ansys Lumerical, "MODE Solutions and INTERCONNECT," 2024. [Online]. https://www.lumerical.com/

[10] M. Köhler, "KLayout layout viewer and editor," 2024. [Online]. https://www.klayout.de/

[11] L. Chrostowski *et al.*, "SiEPIC EBeam PDK," 2024. [Online]. https://github.com/SiEPIC/SiEPIC_EBeam_PDK

[12] "gdsfactory: open-source platform for end-to-end chip design," 2024. [Online]. https://gdsfactory.github.io/gdsfactory/